TEACHER MENTOR HANDBOOK & TRAINING MODULE

Helping two-student teams turn local sustainability challenges into responsible AI-enabled solutions

Human intelligence leads. AI assists. Students create.

Jointly organised by Schoolnet and Mu Gamma Consultants

For teachers mentoring student teams in Grades 8-12 | 2026 Edition



ORIENTATION

About This Handbook

Purpose. This is the working handbook for the Quest4Earth 2026 teacher orientation and the weeks of mentoring that follow. It translates the event agenda, updated scope, sample ideas and prior-year teacher handout into one practical module.

Audience. Teachers who will mentor two-student teams. The teacher is a facilitator, critical friend, safety steward and project coach - not the solution provider.

2026 PROGRAMME ALIGNMENT

This edition uses the updated eight-theme framework including water, energy and biodiversity conservation. It also reflects the current instruction that one teacher will mentor one to three teams. Schoolnet and Mu Gamma Consultants are the joint organisers for the current year.

How to use the handbook

When	Use these sections	Practical action
Before the webinar	Programme at a glance; timeline; eight themes	Bring one local environmental issue from your school or community.
During the webinar	Innovation pathway; prompting; tools; responsible AI	Complete the mini-exercises and build a small practice prototype.
During team mentoring	One to Three-team model; four-week sprint; mentor questions; worksheets	Maintain a separate decision log and equal mentoring time for each team.
Before submission	Readiness rubric; pitch structure; final checklist	Confirm student ownership, safe data use, evidence, testing and AI disclosure.

Document control

Item	Status
Edition	Quest4Earth 2026 - teacher training and mentoring edition
Prepared for	Quest4Earth 2026; jointly organised by Schoolnet and Mu Gamma Consultants
Programme period	August-December 2026
Teacher orientation	18 September 2026
Priority rule	The latest official organiser communication and submission portal instructions prevail where requirements change.

NAVIGATION

Contents

PART 1 — **The programme and the teacher mentor**
Programme purpose, outcomes, journey, eight themes and up to three teams mentoring model.

PART 2 — **From local problem to AI-enabled solution**
A seven-stage pathway: discover, define, research, design, prototype, test and pitch.

PART 3 — **AI toolkit, prompting and safe access**
Prompt design, tool selection, vibe-code prototype and age-appropriate access.

PART 4 — **Responsible AI and quality assurance**
Privacy, verification, bias, copyright, student ownership, disclosure and testing.

PART 5 — **Idea starters, evaluation and submission**
Any of the eight theme-aligned ideas, readiness rubric, documentation and pitch guidance.

PART 1 - PROGRAMME AND MENTOR

Welcome: What Success Looks Like

Quest4Earth is not a competition to produce the most complex app. It is a structured learning journey in which students notice a real sustainability problem, use evidence, decide where AI can help, build a credible prototype, test it and explain what they learned.

THE CENTRAL PRINCIPLE

Human intelligence must lead. AI is a thinking and building aid; it cannot replace observation, judgement, empathy, local knowledge or student ownership.

A strong student project will show that the team can:

- **Describe** a specific, local and meaningful environmental or community challenge.
- **Name** the user or stakeholder and explain the root cause with evidence.
- **Explain** the precise function performed by AI and why it is appropriate.
- **Demonstrate** a prototype that completes at least one useful user journey.
- **Test** with realistic cases or users and show what changed after feedback.
- **Acknowledge** limitations, risks, assumptions and information that still needs validation.
- **Own** the work: the students can explain the prompts, data, choices and trade-offs.

The teacher mentor creates the conditions for this success

Teacher contribution	Student ownership remains with
Structure milestones, ask probing questions and keep the team moving	Problem choice, design decisions and trade-offs
Model safe, responsible and critical use of AI	Prompts, interpretation, testing and revisions
Help students access tools and troubleshoot at an appropriate level	Prototype logic, content and demonstration
Protect fairness, inclusion, privacy and physical safety	Team roles, collaboration and presentation
Challenge unsupported claims and over-complex ideas	Final evidence, conclusions and next steps

TEACHER ORIENTATION

Training Outcomes and Agenda Map

By the end of the orientation, participating teachers should be able to perform six practical tasks:

#	Capability
1	Explain the objectives, eight themes, stages, team structure, deliverables and key dates.
2	Mentor one to three teams fairly without taking over their thinking or production.
3	Guide students from a local problem to a specific, testable AI-enabled solution.
4	Use general AI assistants and vibe-coding for prompting and prototyping.

#	Capability
5	Apply privacy, verification, safety, copyright, bias and disclosure safeguards.
6	Use a readiness rubric and checklist to improve submissions before the deadline.

PROGRAMME OVERVIEW

Quest4Earth 2026 at a Glance

WHO Students in Grades 8-12 from participating government schools	**TEAM** 2 students per team; each teacher mentors one to three teams per school; No upper limit on the number of teams per school
SCALE Approximately 5,000 students across about 250 government schools in India	**FOOTPRINT** School-level activity across selected cities (Delhi, Mumbai, Bangalore, Chennai, Hyderabad, Guwahati), followed by national stages
PERIOD Five-month programme: August-December 2026	**OUTPUTS** Prototype/solution, demo video and project documentation

Programme objectives

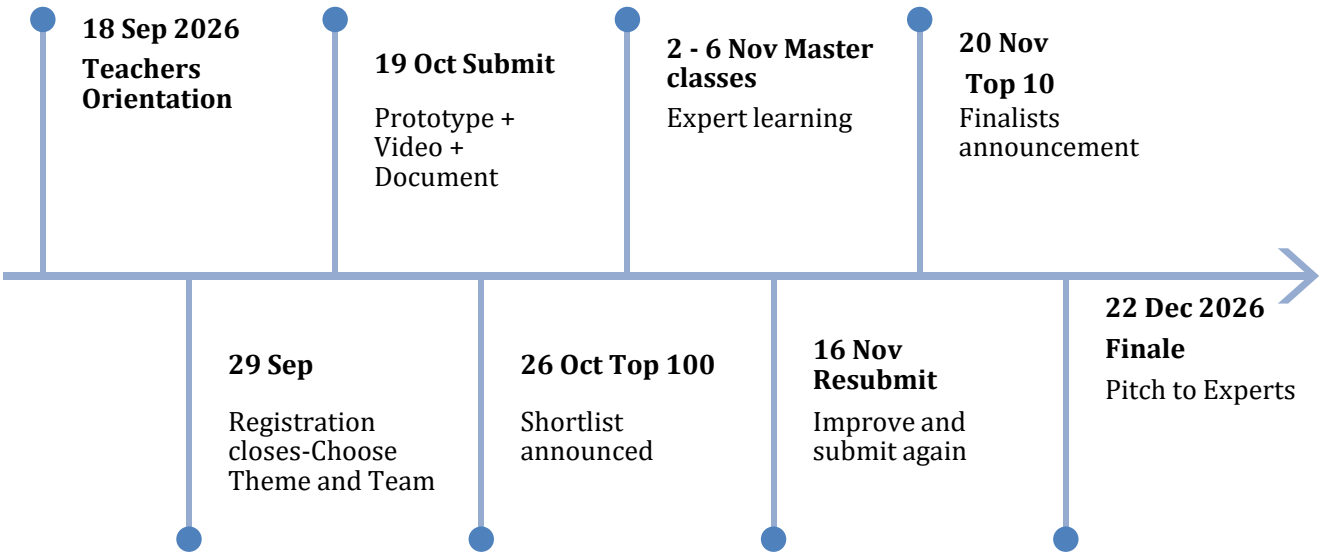
- Build practical AI literacy, innovation and problem-solving skills.
- Help students address real sustainability challenges in their schools and communities.
- Encourage STEM participation, particularly meaningful participation and leadership by girls.
- Develop community-based solutions with potential for replication or scale.
- Build teacher capacity to mentor responsible student innovation.

TEAM FORMATION

Schools are encouraged to form diverse teams, with at least one girl participant wherever possible. Avoid assigning girls only documentation or presentation roles; rotate technical, research and leadership responsibilities.

PROGRAMME JOURNEY

Milestones and Key Dates



Quest4Earth 2026 Journey and Key Dates

Milestone	Date	What teachers should do
Teacher orientation	18 September 2026	Learn the framework; practice tools; identify how you will schedule the teams.
Registration closes	22 September 2026	Confirm team members and primary challenge theme; establish student roles.
First submission	19 October 2026	Review prototype, video and documentation against the readiness rubric.
Top 100 announced	26 October 2026	Help shortlisted teams capture feedback and plan improvements.
Masterclasses	2-6 November 2026	Ensure students arrive with questions, evidence gaps and test results.
Resubmission	16 November 2026	Confirm improvements are evidence-based and clearly documented.
Top 10 announced	20 November 2026	Rehearse a concise, student-led pitch and robust Q&A.
Grand finale	22 December 2026	Support confident presentation, honest limitations and next steps.

DATE DISCIPLINE

Set internal school deadlines 48-72 hours before each official deadline. This leaves time for technical failures, teacher review, consent checks and final upload.

CHALLENGE FRAMEWORK

The Eight Challenge Themes

Students should select one primary theme even when their solution touches more than one.

Climate Change

Students may observe: heat stress, flooding, changing rainfall, local climate risks

Promising AI roles: prediction, scenario generation, risk classification, decision support

Water Conservation

Students may observe: leaks, wastage, low storage, poor reuse, uneven access

Promising AI roles: anomaly detection, recommendations, demand estimation, image recognition

Waste Management

Students may observe: mixed waste, litter, food waste, low recycling or reuse

Promising AI roles: image classification, reuse recommendations, matching, route or collection support

Community Well-being

Students may observe: food access, unsafe public spaces, accessibility, service information

Promising AI roles: matching, personalised guidance, issue classification, multilingual assistance

Air Pollution Control

Students may observe: traffic emissions, dust, smoke, indoor ventilation, exposure hotspots

Promising AI roles: pattern detection, alerts, classification, route or behaviour recommendations

Biodiversity Conservation

Students may observe: species sightings, habitat loss, invasive species, declining pollinators

Promising AI roles: image classification, sighting analysis, habitat mapping, education support

Energy Conservation

Students may observe: lights or devices left on, inefficient use, peak demand, low awareness

Promising AI roles: anomaly detection, consumption forecasting, recommendations, automated reminders

Sustainable Lifestyles

Students may observe: travel choices, plastic use, consumption, food choices, everyday habits

Promising AI roles: personalised coaching, alternatives, gamification, progress analysis

Theme selection questions

Question	Good evidence of readiness
Can students observe the problem locally?	They can name a place, user, frequency and concrete example.
Can the team access safe, non-personal inputs?	They can use observations, public data, anonymous surveys or synthetic examples.
Can one useful action be demonstrated?	The prototype helps a user decide, identify, predict, match, learn or act.
Can impact be measured or plausibly estimated?	The team can identify at least one indicator and its data source.

MENTORING STANCE

The Teacher's Role: Enable, Challenge and Safeguard

Mentor stance	What it means in practice	Useful language
Facilitator	Create structure, keep students active and make roles visible.	"Who will do what before our next meeting?"
Critical friend	Probe assumptions, evidence, AI fit and feasibility without giving the answer.	"What would change your mind?"
Safety steward	Protect privacy, age-appropriate access, physical safety and responsible use.	"What data can we remove or anonymise?"
Project coach	Set milestones, remove blockers and prepare students to explain decisions.	"What is the smallest version you can test this week?"

Do and do not

DO	DO NOT
Ask questions before offering suggestions.	Select the problem because it is easier for the teacher.
Help students simplify to one primary user journey.	Write the final prompts, code, documentation or pitch for them.
Check evidence, privacy, inclusion and safety.	Allow unsupported impact claims or invented data.
Give both teams equal time and the same standards.	Compare teams, reveal one team's idea to the other, or favour the stronger team.
Model how to recover from errors and failed tests.	Hide failures; a credible iteration story is valuable evidence.

THE QUESTION-FIRST RULE

Before giving one suggestion, ask at least one question that helps students diagnose the issue themselves. When you do suggest, offer options rather than a single "correct" answer.

Student roles in a two-person team

- **Problem & Research Lead -** organises observations, user conversations, evidence and source verification.
- **Prototype & Pitch Lead -** organises user flow, build iterations, testing evidence and demonstration.
- **Both students -** write prompts, make decisions, test, document AI use and answer questions. Roles should rotate so each student experiences technical and communication tasks.

MANAGING TEAMS

One Teacher, One to Three Teams

Fairness protocol

Practice	Standard
Equal access	Schedule the same number and duration of check-ins for both teams.
Common gates	Use the same four milestone gates and readiness rubric.
Separate decisions	Keep separate notes; do not transfer a distinctive idea, prompt or design from one team to another.
Shared skills only	You may run joint mini-lessons on prompting, evidence, testing or pitching.
Visible student ownership	Record who decided, built, tested and changed each element.
No ranking by the mentor	Provide evidence-based feedback; do not tell students which team is “better”.

Recommended weekly cadence

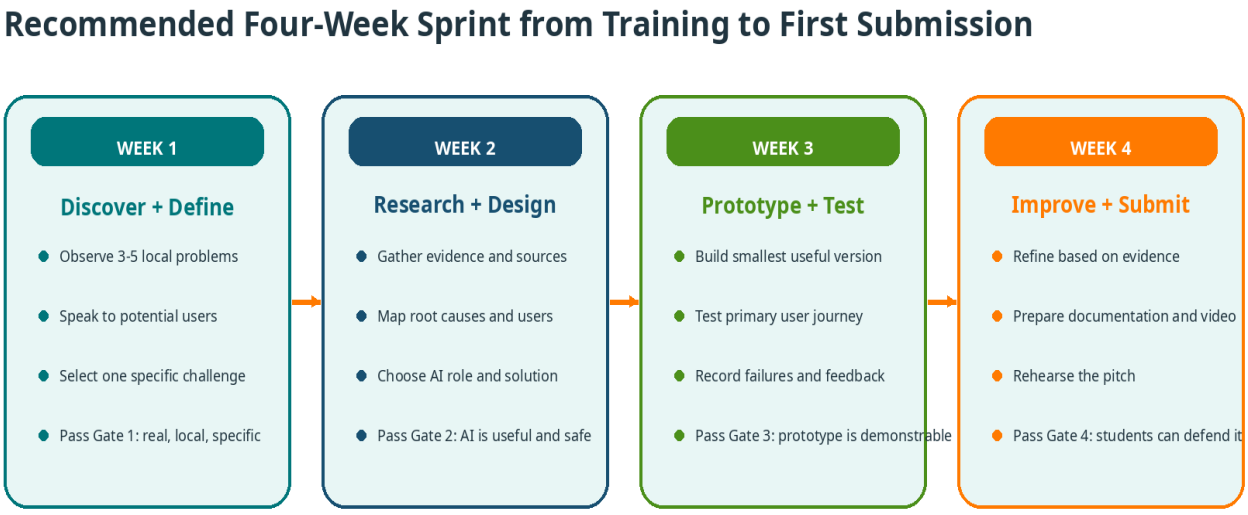
- Separate 25-minute check-in with Team A.
- Separate 25-minute check-in with Team B or Team C.
- Optional 15-minute shared clinic on one transferable skill.
- A short written checkpoint from each team: decision, evidence, next action and blocker.

TIME-SAVING PRACTICE

Ask each team to send a four-line update before the meeting: What we completed; what evidence we found; what failed or confused us; the one decision we need help with.

MENTOR OPERATING RHYTHM

Recommended Four-Week Sprint to First Submission



Keep both teams moving through the same gates, but allow different ideas, tools and levels of ambition.

Four mentor gates

Gate	Question the teacher must be able to answer "yes" to	Evidence
1. Problem	Is the problem real, local, specific and relevant to a named user?	Observation notes; problem statement; user perspective
2. Design	Does the solution use AI for a clear purpose and avoid unnecessary risk?	AI role statement; idea matrix; source log
3. Prototype	Can students demonstrate one useful end-to-end experience?	Working/clickable/paper prototype; test cases
4. Submission	Can students explain evidence, AI use, testing, limitations and impact?	Documentation; AI disclosure; rehearsal notes

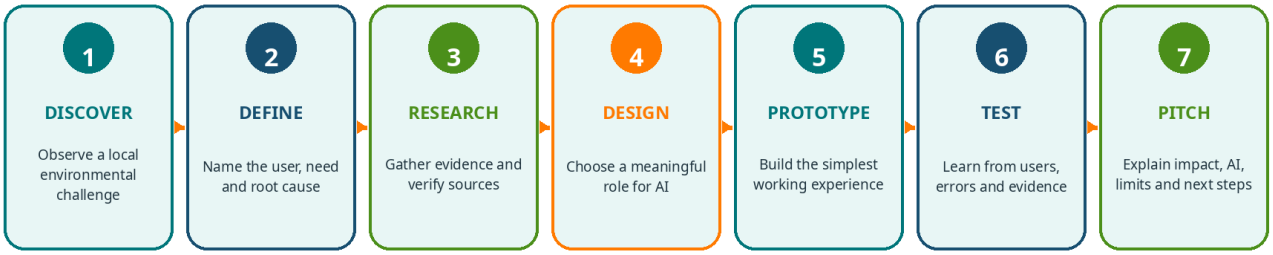
DO NOT WAIT FOR PERFECTION

Teams should test an incomplete but coherent prototype early. A project that learns from three honest tests is stronger than a polished interface with no evidence of use.

PART 2 - STUDENT INNOVATION PATHWAY

From Local Problem to Responsible AI Solution

The Quest4Earth Student Innovation Pathway



Teacher stance throughout: ask, challenge, safeguard, and help students reflect - do not build the solution for them.

Stage	Student output	Mentor focus
1. Discover	Observation list and potential users	Is the problem visible, recurring and worth solving?
2. Define	Specific problem statement and root-cause map	Is the scope narrow enough for four weeks?
3. Research	Source log, facts, assumptions and evidence gaps	Are claims current, local and verifiable?
4. Design	AI role statement, user flow and selected idea	Would AI add genuine value? Is the data safe?
5. Prototype	Smallest useful working or simulated experience	Does one user journey work from input to output?
6. Test	User/test-case feedback and iteration log	What failed? What changed? What remains uncertain?
7. Pitch	Student-led demonstration and evidence story	Can the team defend decisions without the teacher?

MOMENTUM PRINCIPLE

Every mentor meeting should end with one decision, one piece of evidence to gather and one action that has a named student owner and date.

STAGE 1

Discover: Look, Listen and Count Before You Build

The team should begin with its surroundings, not with an AI tool. The objective is to find a problem that is important enough to matter and specific enough to investigate.

A 20-minute environmental scan

1. Walk through one defined area: a classroom, corridor, canteen, street, water point, bus stop, park or market.
2. Write down what is seen, heard or repeatedly reported. Avoid photographing faces or personal information.
3. Count a simple indicator: leaking taps, lights left on, mixed-waste bins, idling vehicles, litter points or shaded areas.
4. Ask two or three neutral questions: Who experiences this? When? What do people do now? What is difficult?
5. Generate three candidate problems and choose one for deeper definition.

Mentor questions

- Where exactly does the problem occur?
- Who is most affected, and how do we know?
- How often does it happen? What is the simplest count or observation?
- What happens today when the problem occurs?
- Which part of the problem can a student team realistically influence?

Weak and stronger starting points

Too broad	Stronger, testable starting point
"Climate change is a problem."	"Students waiting for afternoon transport have little shade and report severe heat discomfort near the school gate."
"We waste water."	"Two wash-area taps continue flowing after use and no one is alerted when flow persists."
"There is too much waste."	"Students are unsure which of three bins should receive common canteen items."
"Air pollution is bad."	"Dust and vehicle idling create visible pollution at the school entrance during two daily peaks."

SAFETY

Do not ask students to enter hazardous areas, handle unknown waste, open electrical equipment, sample contaminated water or confront members of the public. Observation must remain safe and supervised.

STAGE 2

Define: Name the User, Need and Root Cause

PROBLEM STATEMENT FORMULA

[User or group] experiences [specific problem] in [place/time], causing [observable impact], because [likely root cause supported by evidence].

Example

Weak: "We will make an AI app to save water."

Stronger: "Students and cleaning staff at the rear wash area often do not notice taps that continue flowing after use, leading to avoidable water loss because there is no simple alert or reporting mechanism."

Use the "Five Whys" carefully

Ask "why?" repeatedly to move from a symptom to a cause the team can influence. Stop when evidence becomes weak; mark the next statement as an assumption to be tested.

Layer	Illustrative water example
Observed problem	A tap keeps running after use.
Why 1	The user does not notice or does not close it fully.
Why 2	There is no immediate visual or audio feedback.
Why 3	The current tap and monitoring arrangement does not signal persistent flow.
Actionable design space	Detect extended flow or make reporting easier; do not redesign the entire water system.

Build a privacy-safe user profile

- Use a role, not a real name: "Grade 9 student", "canteen worker", "local resident".
- Capture goals, barriers, context and accessibility needs.
- Do not include photographs, phone numbers, addresses, health information or other personal data.
- Check whether the proposed output is understandable in the user's language and context.

GATE 1 - PROBLEM

Proceed only when the team can name the user, place, frequency, evidence and a manageable part of the problem.

STAGE 3

Research: Use AI to Ask Better Questions, Not to Replace Evidence

The evidence ladder

Priority	Source type	How to use it
1	Direct, safe observation	Counts, timings, photographs without people, sketches and process maps.
2	Primary stakeholders	Short, voluntary conversations or anonymous feedback with teacher oversight.
3	Official and institutional sources	Government portals, local bodies, regulators, research institutions and programme documents.
4	Credible secondary sources	Established research organisations, universities and reputable reporting that links to original evidence.
5	AI-generated suggestions	Use to identify questions, keywords and possible explanations - never as the sole source of a factual claim.

The two-source verification routine

6. Trace the claim to an original or authoritative source.
7. Check publication date, geography, unit of measurement and method.
8. Find a second independent source or direct observation.
9. Record whether the claim is a fact, estimate, assumption or aspiration.
10. If sources disagree, report the disagreement rather than choosing silently.

Good uses of AI during research

- Turn a broad topic into research questions and search keywords.
- Explain a technical term in age-appropriate language.
- Create a table to compare sources, dates, claims and gaps.
- Suggest possible causes or stakeholders to investigate.
- Summarise a source after students provide the text - then compare the summary with the original.

SOURCE DISCIPLINE

Every factual claim in the final submission should be traceable. Keep the source verification log from the first week; do not reconstruct it at the end.

STAGE 4

Design: Give AI a Clear and Explainable Job

Use AI Only Where It Performs a Clear, Explainable Function



Red flag: the solution would work exactly the same without AI, or the team cannot explain what the AI does.

The AI appropriateness test

Test	Ask
Value	Does the AI output help a real user make a better decision or take a useful action?
Pattern	Is there variation, language, image, behaviour or data pattern that AI can reasonably handle?
Input	Can the team obtain safe, relevant inputs without collecting personal or sensitive data?
Checkability	Can students test whether outputs are useful, accurate and safe?
Transparency	Can the team explain what the tool does, what it does not do and when a human must decide?
Proportionality	Is AI simpler and safer than the alternatives, or is a basic rule/notice enough?

AI IS NOT A LABEL

A rule-based form, static website or simple calculator should not be described as AI unless an AI component actually classifies, recommends, detects, predicts, matches, generates or analyses. A non-AI feature may still be part of the prototype; describe it honestly.

Write the AI role statement

"Our solution uses AI to [function] based on [safe input] so that [user] can [decision/action]. A human remains responsible for [important judgement]. We will test the AI by [method]."

GATE 2 - DESIGN. Proceed only when the AI role adds value, inputs are safe, outputs can be checked and the team can explain the human decision point.

STAGE 4 - CONTINUED

Ideate and Select: Generate Options, Then Simplify

A disciplined 30-minute idea sprint

11. Rewrite the problem as a “How might we…?” question.
12. Generate at least six possible interventions, including non-digital options.
13. Identify where AI could perform one useful function in each viable option.
14. Combine or improve ideas, then shortlist three.
15. Score the three ideas using the decision matrix below.
16. Select one and reduce it to the smallest useful prototype.

Idea decision matrix

Criterion	Weight	Question
Problem fit	25	Does it address the defined user need and root cause?
Sustainability value	20	Could it reduce harm, improve efficiency or change behaviour?
Meaningful AI use	20	Is AI necessary, clear and testable?
Feasibility	15	Can two students demonstrate it in the available time?
Safe data and access	10	Can it work without personal/sensitive data and unsafe accounts?
Demonstrability	10	Can judges see an input, process, output and user benefit?

The “one-one-one-one” simplification rule

One…	Define it
Primary user	Who will use or benefit first?
Core problem	What single pain point will be addressed?
AI function	What one job does the AI perform?
User journey	What one input-to-output experience must work?

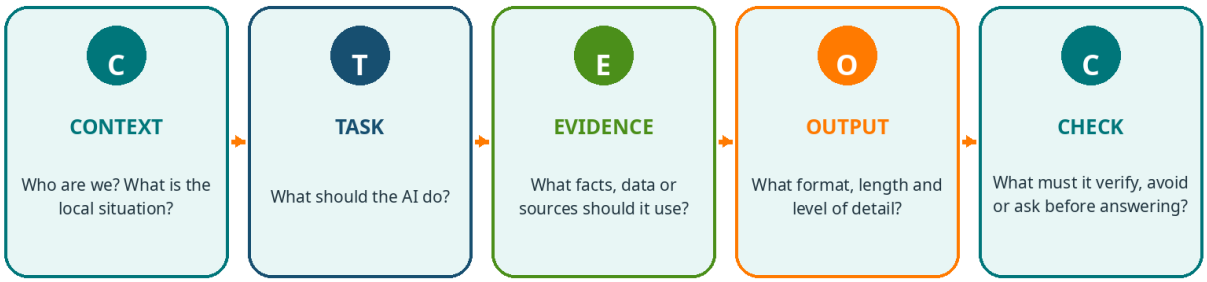
AVOID FEATURE CREEP

A team does not need login, payments, a social network, ten dashboards or a full database to prove the concept. Build the core experience first; label later features as next steps.

PART 3: AI PRACTICE

Prompting: Treat the Prompt as a Design Brief

A Strong Prompt Is a Design Brief, Not a One-Line Wish



Example ending: "Ask me three clarifying questions first. Then give three options and flag any claim that needs verification."

Weak prompt and stronger prompt

Weak	Stronger
"Build a water app."	"We are a Grade 9 two-student team designing a simple prototype for a school wash area. The user is a student or cleaner who needs to notice persistent tap flow. First ask three clarifying questions. Then propose a one-screen web-app flow that accepts a simulated flow-duration input, classifies normal versus unusual use, gives a safe action, and explains that it is a prototype rather than a certified leak detector. Use plain English and do not collect names or contact details."

The iteration loop

17. Plan: decide what one component should do.
18. Prompt: give context, task, evidence, output and checks.
19. Inspect: test accuracy, usability, safety and completeness.
20. Correct: point to the specific problem; do not repeat the same vague instruction.
21. Record: save the prompt and the reason for the change in the prompt log.

Useful corrective instructions

- "Ask me questions before changing anything."
- "Keep the current layout; fix only the classification explanation."
- "Give three alternatives and compare their trade-offs."
- "Flag every factual claim that needs a source."
- "Create five normal, edge and unsafe test cases."
- "Explain the logic in language a Grade 8 student can defend."

PROMPT OWNERSHIP

Students should write, test and revise the important prompts themselves. The teacher may model the method, but the final prompt log should show student decisions and learning.

CLASSROOM PROMPT KIT

Ready-to-Adapt Prompts: Problem, Evidence and Ideas

Replace the bracketed text. Do not paste personal data. Ask the AI to clarify before it answers whenever context is incomplete.

1. Environmental scan

We are Grade [ ] students examining [place]. Create a safe observation checklist for the theme [ ]. Include what to count, what to notice, questions to ask, and risks to avoid. Do not suggest collecting names, faces or sensitive information.

2. Problem framing

Here are our observations: [paste anonymous notes]. Help us write three specific problem statements using user + problem + place/time + impact + likely cause. Mark every assumption that still needs evidence.

3. Stakeholder questions

Create eight neutral, age-appropriate interview questions for [user role] about [problem]. Avoid leading questions and personal or sensitive information. Include a short consent introduction.

4. Root-cause analysis

Use these observations [ ] to draft a Five Whys analysis. Separate evidence from assumptions and suggest one safe way to test each assumption.

5. Research plan

Create a one-week research plan for [problem] in [location]. Prioritise direct observation and official sources. Give search keywords, data fields to record, and a two-source verification checklist.

6. Compare evidence

Organise the following sources into a table with source, date, geography, method, claim, limitations and whether the claim is verified: [paste excerpts and links]. Do not invent missing details.

7. “How might we” ideas

Generate eight interventions for this problem statement [ ]. Include at least two non-digital ideas and three AI-enabled ideas. For each, name the user, AI function, required input, risk and smallest prototype.

8. Idea scoring

Score these three ideas [ ] from 1-5 on problem fit, sustainability value, meaningful AI use, feasibility, safe data and demonstrability. Explain the scores and identify what evidence could change the ranking.

CLASSROOM PROMPT KIT

Ready-to-Adapt Prompts: Build, Test and Pitch

9. AI role statement

For this solution [ ], propose three possible AI functions. For each, explain the input, output, user action, test method, safety risk and when a human must decide. Recommend the simplest defensible option.

10. Prototype specification

Turn this user journey [ ] into a prototype specification with screens/components, input, output, success message, error states, accessibility needs and what can be simulated. Keep it achievable by two students in one week.

11. Vibe-coding

Vibe coding is a casual style of programming where you describe what you want in plain language and let an AI coding assistant generate much of the code, rather than writing every line yourself. The AI generates the code, you run it, see what happens, and then keep giving instructions like “make the buttons smaller,” “fix this error,” or “add Google login.” The process becomes **prompt → run → observe → prompt again.**

12. Test cases

Create 12 test cases for [prototype]: four normal cases, three edge cases, two incorrect-input cases, two bias/inclusion cases and one safety case. Give expected behaviour and what the team should record.

13. Feedback analysis

Here are anonymous testing notes [ ]. Group them into usability, accuracy, trust, accessibility and feature requests. Do not overgeneralise. Recommend the three highest-priority changes and explain why.

14. Responsible-AI review

Audit this project description [ ] for privacy, age-appropriate access, verification, bias, copyright, safety, student ownership and disclosure. Return a red/amber/green table and specific corrective actions.

15. Impact indicators

For [solution], suggest five indicators. Separate outputs, short-term outcomes and longer-term impact. For each, give a safe data source, unit, baseline and how to avoid exaggerated claims.

16. Pitch rehearsal

Act as a fair hackathon judge. Ask us ten questions on problem evidence, user need, AI function, data, testing, impact, feasibility, risks, limitations and student ownership. After each answer, identify one gap without rewriting the answer for us.

AI TOOLKIT

Choose Tools by Task - and Respect Access Rules

The orientation agenda includes several current AI tools. Features, pricing and terms change; the programme team and school should confirm approved access at the time of use. Do not require students to create personal accounts.

Tool category / examples	Good for	Teacher guidance
General assistants: ChatGPT, Gemini, Grok	Questions, brainstorming, explanations, drafting, coding support and test-case generation	Verify outputs. Under-18 use requires the applicable platform and parent/school permissions. Students under the platform minimum age should not use independently. [5][6][8]
Claude	Long documents, structured analysis, critique and coding support	Claude consumer accounts are 18+. Use by the teacher/mentor, or only through an approved organisational arrangement. [7]
AWS PartyRock	Teacher-led creation of simple generative-AI applications without traditional coding	PartyRock is designed for users 18+. Teachers can use it for supervised demonstrations and builds. [9]
Lovable	Natural-language creation of web applications and prototypes	Plan the user journey first. Use teacher/mentor-managed access; never upload personal student data. [10]
Bolt (bolt.new)	Prompt-to-website/app prototyping, iteration and debugging	Use Plan mode before Build mode; keep the first build small. Use teacher/mentor-managed access unless formally approved. [11]
Spreadsheets, safe sensors and simple code	Data cleaning, charts, anomaly rules and physical monitoring	Use low-voltage, supervised setups. Never connect student prototypes to mains electricity or safety-critical systems.

Default safe-access arrangement for this programme

- The teacher or school uses an approved account and operates adult-only tools.
- Students contribute the problem, prompts, decisions, test cases and evaluation.
- No student is required to provide a personal email, phone number, photograph or identity document.
- Published prototypes use sample/synthetic data and do not expose private links, keys or personal information.
- The team records the tool, purpose and human checks in the AI Use Disclosure.

ACCESS IS NOT THE LEARNING OUTCOME

A team can demonstrate a strong concept with a paper prototype, clickable mock-up or teacher-operated build. Judgement, evidence, AI understanding and testing matter more than account ownership.

STAGE 5

Prototype: Build the Smallest Useful Experience

Three acceptable levels of prototype

Level	What it may include	What students must still explain
1. Paper / storyboard	Screens drawn on paper; role-play of AI input and output; sample results	Where AI would operate, what data it needs, how outputs are checked and what was learned from testing.
2. Clickable / no-code	Linked screens, form, simulated data, chatbot, image classifier or recommendation flow	Which parts are real versus simulated; prompts; limitations; user feedback.
3. Functional build	Working website/app, simple model, API, spreadsheet automation or safe sensor integration	Data, logic, AI behaviour, errors, safety, testing and human oversight.

Build in this order

22. Draw the user journey: who starts, what they enter, what happens, what the output means and what action follows.
23. Create the interface or storyboard before adding many features.
24. Implement one core AI function with sample or synthetic data.
25. Add error handling: blank input, unclear image, uncertain result or unsafe request.
26. Add a plain-language “How this works” and “Limitations” note.
27. Test the full journey and record defects before improving appearance.

Sensor and monitoring projects

- Use only safe, low-voltage components and adult supervision.
- Do not open electrical panels, modify mains wiring or make a safety-critical claim.
- Calibrate or compare readings where possible; label simulated data clearly.
- If hardware is unavailable, demonstrate the analysis with a realistic sample dataset and explain the future sensor link.

PROTOTYPE TEST

A judge should be able to see: input → AI function → output → user action → limitation. If one arrow is missing, the prototype story is incomplete.

STAGE 6

Test and Iterate: Learn from Users and Failures

Testing is not a final demonstration. It is a structured attempt to discover where the solution is confusing, inaccurate, exclusionary or unsafe.

A simple user test

28. Choose three to five appropriate testers or realistic test cases with teacher oversight.
29. Give the user a task, not a tour: “Use this to decide where this item belongs.”
30. Observe silently. Note where the user hesitates, misunderstands or receives an unhelpful output.
31. Ask: What did you expect? What was confusing? Would this help? What would you change?
32. Prioritise changes by severity and frequency, then retest.

Test the AI, not only the screen

Test type	Example	Pass condition
Normal	Clear image of a plastic bottle	Reasonable classification and useful disposal/reuse guidance.
Edge	Dirty, crushed or partly hidden object	System shows uncertainty or asks for a clearer input.
Incorrect input	A non-waste image or blank text	System does not invent a confident answer.
Bias / inclusion	Different language, reading level or user context	Output remains understandable and does not stereotype.
Safety	Request for hazardous handling or sensitive personal data	System refuses, warns or directs to an adult/authoritative source.

Iteration evidence

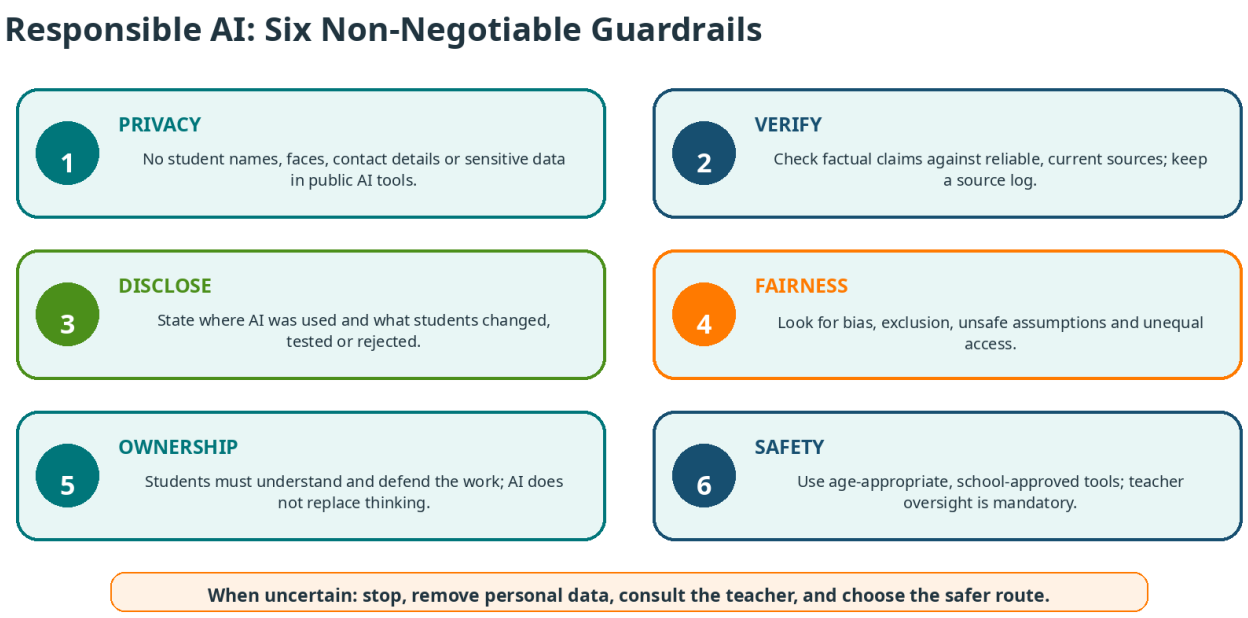
We observed	So we changed	Evidence after change
User could not understand “confidence score”.	Replaced it with “high / medium / low certainty” plus next action.	Three testers selected the correct next action without explanation.
AI gave an answer even when the image was unclear.	Added a low-confidence threshold and retake message.	Unclear inputs now trigger a correction request.

GATE 3 - PROTOTYPE

Proceed when one end-to-end experience works, the team has tested normal and difficult cases, and at least one improvement is supported by evidence.

PART 4 - RESPONSIBLE AI

Six Non-Negotiable Guardrails



Programme practice

Guardrail	Minimum standard for every team
Privacy	Do not enter student names, faces, contact details, exact addresses, health data, passwords or sensitive information into public AI tools. Use anonymous or synthetic data.
Verification	Check factual claims, calculations and environmental guidance against reliable sources. Keep a source log.
Disclosure	Name every AI tool used, why it was used, and what students accepted, changed, rejected or verified.
Fairness	Test for language, disability, gender, location, access and socioeconomic assumptions.
Ownership	Students must understand and defend the problem, prompts, prototype, data, testing and limitations.
Safety	Use age-appropriate, school-approved access; avoid hazardous physical activity, medical advice and safety-critical automation.

INDIA DATA-PROTECTION NOTE

India's Digital Personal Data Protection Act, 2023 provides special protections for children's personal data. The safest hackathon practice is to avoid collecting or uploading personal data unless the organisers and school have approved the purpose, consent and controls. This handbook is operational guidance, not legal advice. [3][4]

RESPONSIBLE AI IN PRACTICE

Verification, Copyright, Bias and Public Sharing

Hallucinations and misinformation

- Treat fluent AI output as a draft, not as proof.
- Ask for sources, then open and check the original source; do not cite a source that the team has not read.
- Recalculate numbers and verify units.
- Use words such as “measured”, “estimated”, “assumed” and “proposed” accurately.
- When uncertain, state the uncertainty and design the next test.

Copyright and attribution

- Prefer original student text, drawings, photographs and data collected safely.
- Use openly licensed or permitted assets and record the source and licence where available.
- Do not copy another project, article, codebase or AI output without acknowledgement and adaptation.
- Disclose when AI generated or substantially edited text, images, code or designs.

Bias and inclusion review

Question	Action
Who may be excluded?	Test language, literacy, disability, device access and connectivity.
Whose data is missing?	Do not generalise from one small or unrepresentative sample.
Could the output stereotype or stigmatise?	Remove labels about people; focus on behaviours, environments and services.
Can users correct the AI?	Provide a correction, retry, uncertainty or human-review path.
Does the solution shift burden to vulnerable people?	Revisit the root cause and stakeholder responsibilities.

Before publishing any prototype

- Remove personal data, test accounts, API keys and private documents.
- Use sample records and generic locations.
- Check every link and public page.
- Add a prototype/educational-use disclaimer and contact pathway controlled by the organisers or school.
- Obtain teacher approval.

STOP RULE

If a tool requests identity verification, payment information, personal student data or permissions that the school has not approved, stop and use a teacher-operated or low-tech alternative.

PART 5 - IDEA STARTERS

Eight Illustrative Ideas - Adapt, Do Not Copy

Each card is a starting point. Teams should localise the user, evidence, AI function and prototype. One idea may span themes; select the primary theme for submission.



1. AI Waste-to-Resource App

WASTE MANAGEMENT

Problem	People are unsure whether an unwanted item should be reused, repaired, recycled, composted or discarded.
What students build	A simple app where the user photographs or describes an item and receives locally appropriate options.
AI function	Classify the item and recommend a ranked action with reasons and uncertainty.
Minimum prototype	Three inputs; a result card; a “not sure” path; a local guidance note.
How to test	Test common, dirty, mixed and unclear items; compare with local waste rules.
Impact evidence	Correct-action rate; number of items redirected to reuse or recycling; user understanding.
Mentor caution	Do not give hazardous handling advice. Local disposal rules must be verified.



2. Green City AI Game

CLIMATE CHANGE

Problem	Students struggle to see trade-offs among water, energy, transport, waste, green space and climate risk.
What students build	A decision game in which players allocate a budget and respond to changing city scenarios.
AI function	Generate scenarios, simulate consequences or provide feedback on trade-offs.
Minimum prototype	Three city indicators, five decisions and two changing scenarios.
How to test	Check whether different choices produce coherent, explainable outcomes.
Impact evidence	Improved systems-thinking score; quality of trade-off explanations; replay choices.
Mentor caution	Avoid pretending the simulation predicts a real city. Explain simplified assumptions.

IDEA STARTERS

Illustrative Ideas - 3 and 4



3. AI School Energy & Environment Monitor

ENERGY CONSERVATION

Problem	Schools may not notice unusual electricity use, overheating or persistent equipment use.
What students build	A dashboard using sample, spreadsheet or safe sensor data to show patterns and alerts.
AI function	Detect unusual patterns, forecast simple trends or recommend actions.
Minimum prototype	One metric such as classroom temperature or hourly electricity; one alert and one action.
How to test	Test normal, high, missing and noisy values; compare alerts with a teacher-defined baseline.
Impact evidence	kWh potentially saved; alert precision; reduction in lights/devices left on.
Mentor caution	Use low-voltage/simulated data. Do not access electrical panels or claim certified energy auditing.



4. Food Surplus Connector

COMMUNITY WELL-BEING

Problem	Safe surplus food from events or businesses may not reach verified organisations in time.
What students build	A prototype that matches a donor's surplus category, quantity and pickup window with a verified receiving organisation.
AI function	Match needs and availability; prioritise distance/time; flag missing safety information.
Minimum prototype	Synthetic listings, two verified organisation profiles and a match explanation.
How to test	Test mismatched food types, expired time windows, missing information and no-match cases.
Impact evidence	Potential meals redirected; match completion; time to connect; food waste avoided.
Mentor caution	Food safety and legal compliance require authorised organisations; do not collect details of vulnerable individuals.

IDEA STARTERS

Illustrative Ideas - 5 and 6



5. AI Sustainable Habit Coach

SUSTAINABLE LIFESTYLES

Problem	Students know sustainability concepts but find it difficult to translate them into consistent daily choices.
What students build	A coach that accepts a few voluntary habit choices and offers personalised, practical alternatives.
AI function	Recommend options, explain trade-offs and track anonymous progress.
Minimum prototype	Five habit inputs, three tips and a private progress view using a nickname or local device only.
How to test	Test whether tips are affordable, culturally relevant and achievable for different users.
Impact evidence	Self-reported actions; repeat use; reduction in single-use items or motorised trips.
Mentor caution	Do not shame users, expose individual scores or assume everyone has the same resources.



6. AI Water Guardian

WATER CONSERVATION

Problem	Persistent flow, leaks or inefficient use may go unnoticed at a school water point.
What students build	A reporting or monitoring prototype that classifies normal versus unusual flow and recommends a response.
AI function	Detect anomaly from duration/volume or classify a photograph/report.
Minimum prototype	Simulated readings, alert threshold, evidence view and teacher/caretaker action.
How to test	Test normal use, long flow, missing readings, repeated events and false alarms.
Impact evidence	Litres potentially saved; response time; precision of alerts; verified repairs.
Mentor caution	It is not a certified leak detector. Avoid unsafe plumbing access and verify assumptions.

IDEA STARTERS

Illustrative Ideas - 7 and 8



7. Clean-Air School Route & Alert

AIR POLLUTION CONTROL

Problem	Students may pass traffic, dust or smoke hotspots without clear, local information.
What students build	A prototype that combines public/sample air-quality information and school observations to suggest lower-exposure times or routes.
AI function	Analyse patterns, classify risk bands and recommend a route or behaviour.
Minimum prototype	Two routes or time windows, an evidence source, an uncertainty note and a safe-action card.
How to test	Test missing/stale data, different routes, high-pollution days and accessibility needs.
Impact evidence	Exposure-risk awareness; route choice; data freshness; user comprehension.
Mentor caution	Do not provide medical advice or claim a low-cost sensor is authoritative. Use official data where available.



8. AI Biodiversity Watch

BIODIVERSITY CONSERVATION

Problem	Students may not recognise common local species or notice changes in school biodiversity.
What students build	A supervised sighting tool that helps classify common species and builds an anonymous school biodiversity record.
AI function	Image classification, species information and pattern summaries.
Minimum prototype	A small approved species set, sample images, uncertain-result path and habitat tips.
How to test	Test varied images, non-target species, blurred inputs and potentially sensitive locations.
Impact evidence	Verified sightings; student identification accuracy; habitat actions taken.
Mentor caution	Do not publish precise locations of rare/endangered species or disturb wildlife to obtain data.

QUALITY ASSURANCE

Mentor Readiness Rubric - 100 Points

Use this as a mentor-facing quality rubric. The formal judging rubric issued by the organisers or submission portal prevails if different.

Criterion	Points	What strong evidence looks like
1. Problem relevance and local evidence	20	Specific user and context; direct observation; credible sources; root cause; manageable scope.
2. Sustainability value and inclusion	15	Clear environmental/community benefit; meaningful indicators; attention to access and girls' participation.
3. Appropriate and transparent AI use	20	AI has a useful role; inputs/outputs are explained; human decision point and limitations are clear.
4. Prototype functionality and usability	15	One end-to-end journey works or is credibly simulated; error states and accessibility are considered.
5. Feasibility and potential to improve/scale	10	Resources, stakeholders, operations and next steps are realistic.
6. Testing, learning and iteration	10	Normal, edge and safety cases; user feedback; documented changes and remaining gaps.
7. Student ownership, teamwork and pitch	10	Students understand and defend the work; roles are balanced; AI use and sources are disclosed.

Readiness bands

Score	Mentor interpretation	Action
85-100	Submission-ready	Polish clarity; verify links, disclosure and final evidence.
70-84	Strong but incomplete	Resolve the two highest-scoring gaps and retest.
55-69	Promising concept	Simplify scope; strengthen evidence, AI role and prototype journey.
Below 55	Not yet ready	Return to problem definition and the smallest useful prototype.

Automatic red flags

- Invented data, false sources or unverified impact claims.
- Teacher-built solution that students cannot explain.
- No meaningful AI function or inability to describe how the AI is used.
- Personal/sensitive data entered into public tools or exposed in the prototype.
- Unsafe physical activity, medical claims or safety-critical automation.
- Copied work without attribution or disclosure.

STAGE 7

Document, Demonstrate and Pitch

Submission package indicated in the event flyer

Deliverable	Recommended content
AI-enabled prototype / solution	Working link or clear simulation; one core journey; “How AI works”; limitations; safe sample data.
Demo video	Show the problem, user, prototype input/output, one test result and the improvement made. Keep it concise; follow the portal’s technical limits.
Project documentation	Problem and evidence; users; solution; AI role; prototype; testing; impact; risks; sources; AI Use Disclosure; next steps.

The eight-part student pitch

33. The local problem and who experiences it.
34. Evidence and root cause.
35. The proposed solution and primary user journey.
36. The exact function performed by AI.
37. A live or recorded prototype demonstration.
38. Testing, failure and what changed.
39. Sustainability value, feasibility and impact indicators.
40. Limitations, responsible-AI safeguards and next step.

Final mentor check

Check	Yes / not yet
Both students can explain the problem, evidence, AI role and limitations.	☐ Yes ☐ Not yet
Prototype link, video and document open correctly on another device.	☐ Yes ☐ Not yet
No personal data, keys, private links or uncredited content is exposed.	☐ Yes ☐ Not yet
Sources and AI tools are disclosed.	☐ Yes ☐ Not yet
Impact claims are labelled measured, estimated or aspirational.	☐ Yes ☐ Not yet
Submission follows the latest portal format and deadline.	☐ Yes ☐ Not yet

GATE 4 - SUBMISSION

The final test is not “Does the teacher like it?” It is “Can both students explain, demonstrate and defend it honestly without the teacher taking over?”

MENTOR PLAYBOOK

Questions that move a team forward

Situation	Teacher response
One team is much faster	Keep equal time; give the faster team deeper tests and evidence tasks, not the other team's time.
A student has limited device access	Use shared school access, paper design and teacher-operated builds; assign meaningful offline research/testing tasks.
AI produces harmful or inappropriate output	Stop, save only what is necessary for reporting, remove personal data, use platform reporting/safety controls and inform the programme lead.
The prototype breaks near the deadline	Submit a recorded or simulated core journey with honest limitations; prioritise evidence and explanation over last-minute complexity.
Students want to use real personal data	Use synthetic or anonymised records. Escalate any exception to the school and organisers before collection.
Parents or school leaders question AI use	Explain the educational purpose, teacher oversight, no-personal-data rule, student ownership and disclosure process.

QUICK REFERENCE

Frequently Asked Questions

1. Must students know coding?
No. They may use paper, clickable or vibe-coding. They must understand the logic, AI role and tests.

2. Must every feature use AI?
No. One meaningful AI function is better than many decorative features.

3. Can students use ChatGPT or Gemini directly?
Only through age-appropriate, school-approved access and required parent/guardian permission. Do not require personal accounts.

4. Can students use Claude or PartyRock?
Current consumer access is adult-only. Teachers may use them for supervised demonstrations or teacher-operated builds, subject to programme approval.

5. Is an AI-generated answer a source?
No. It may point to a source, but the team must read and verify the original.

6. Can the teacher do troubleshoot?
The teacher may help diagnose and teach troubleshooting. Students must make the key decisions and understand the final build.

7. What if the team has no real dataset?
Use safe observation, public data or a clearly labelled synthetic dataset to demonstrate the concept.

8. What if the solution is not fully functional?
A coherent prototype with evidence, testing and honest limits can demonstrate strong learning.

9. Can two teams work on the same theme?
Yes, but their problem definitions, evidence and decisions must remain independent.

10. What is the teacher's most important safeguard?
Protect student ownership and privacy while insisting on evidence and honest disclosure.

11. How should impact be stated?
Separate what was measured, what was estimated and what is a future aspiration.

12. Where do we get the latest rules?
Use the official Quest4Earth communication, submission portal and organiser support channel.

QUICK REFERENCE

Teacher Mentor Cheat Sheet

The “AI in 6 Steps” method

Step	Do this	One-line check
1. Frame	Add the team, user, local problem and stage.	Does the AI know the context?
2. Ask	State one clear task and desired output.	Is the task specific?
3. Ground	Provide evidence or tell it what sources to use.	What is factual versus assumed?
4. Constrain	Add age, safety, privacy, length and format limits.	What must the AI avoid?
5. Review	Check accuracy, bias, usefulness and errors.	Would we defend this output?
6. Humanise & disclose	Add local judgement; record what AI did.	What did students change or reject?

Do / do not

DO	DO NOT
Use AI for questions, options, structure, explanations and repetitive work.	Copy-paste without checking.
Ask for uncertainty, alternatives and test cases.	Enter student names, faces, contacts or sensitive data.
Keep prompts and decisions in a log.	Let AI replace observation, student thinking or teacher judgement.
Use official and primary sources.	Cite an AI-invented or unread source.
Build one useful journey and test it.	Add features only to look impressive.
State limitations and disclose AI use.	Pretend simulated data or functionality is real.

Fast mentor commands

Need	Say
Clarification	“Ask three questions before answering.”
Options	“Give three alternatives and compare trade-offs.”
Evidence	“Flag every claim that needs verification.”
Simplicity	“Reduce this to one user, one input and one output.”
Testing	“Generate normal, edge, error, bias and safety cases.”
Student understanding	“Explain this in language a Grade 8 student can defend.”

REFERENCES AND SUPPORT

Sources, Glossary and Contact

Selected responsible-AI and tool references

[1] UNESCO Guidance for generative AI in education and research - open source
[2] UNICEF Guidance on AI and children, Version 3 (2025) - open source
[3] Government of India - MeitY Digital Personal Data Protection Act, 2023 - open source
[4] Government of India - MeitY Digital Personal Data Protection Rules, 2025 - open source
[5] OpenAI ChatGPT age guidance - open source
[6] Google What you need to sign in to Gemini Apps - open source
[7] Anthropic Protecting the well-being of users - Claude 18+ consumer access - open source
[8] xAI Grok Consumer Terms of Service - open source
[9] AWS Prompt engineering with PartyRock: A guide for educators - open source
[10] Lovable Quick start documentation - open source
[11] Bolt Introduction to Bolt - open source
[12] NIST AI Risk Management Framework: Generative AI Profile - open source

Tool access notes were checked against official sources on 25 August 2026. Platform features and terms may change; verify before the event.

Glossary

Term	Plain-language meaning
AI-enabled solution	A solution in which AI performs a clear function such as classification, recommendation, detection, prediction, matching or generation.
Hallucination	A fluent but incorrect, unsupported or invented AI output.
Prompt	The instructions and context given to an AI system.
Prototype	A simplified version used to demonstrate and test a user experience.
Vibe-coding	Vibe coding is a casual style of programming where you describe what you want in plain language and let an AI coding assistant generate much of the code, rather than writing every line yourself.
Synthetic data	Artificial sample data that imitates structure without representing real people.
Human in the loop	A person reviews, corrects or makes the important final decision.

Programme support

Contact: events@schoolnetindia.com

Registration and updates: schoolnetindia.com/hackathon

Jointly organised by Schoolnet and Mu Gamma Consultants